

Introduction to Microeconomics

11–Profit Maximization, Supply and Market Equilibrium in Perfectly Competitive Markets

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Perfectly Competitive Market

A perfectly competitive market is characterized by

- Many small firms compared to the total size of the market.
- Homogeneous product.
- Full information.
- Negligible Transaction Costs.
- Free Entry and Exit.

We will develop the theory of the firm operating in a perfectly competitive market. However, many of the insights that we develop here extends to cases in which some of the characteristics above are not true. Turns out that none of the above are essential for a market to be “competitive”. What is important is that the market has a **contestable structure: low or nonexistent sunk costs, and the possibility to use strategies to undercut rivals, i.e., price adjustments by the firms are felt strongly by consumers.**

Revenues

Given a market demand curve $Q = D(p)$ with inverse demand curve $p = D^{-1}(Q)$, if the firm faces this demand curve (in full) revenues for the firm are defined as:

$$R(Q) = \text{Price} \times \text{Quantity} = D^{-1}(Q) \times Q.$$

For example, if $Q = a - Bp$, the inverse demand is $p = a/b - Q/b$, thus

$$R(Q) = \left(\frac{a}{b} - \frac{Q}{b}\right)Q$$

Profit Maximization

The firm's profit function, for any q , is

$$\Pi(q) = R(q) - C(q)$$

where $R(q)$ are revenues and $C(q)$ are costs as a function of quantity.

The firm's problem is then to **maximize profits** by choosing optimally the quantity offered.

$$\max_q R(q) - C(q)$$

There are two steps in this problem:

- Output decision. What output level q^* maximizes profit or minimizes loss?
- Shutdown decision. Is it more profitable to produce q^* or to shutdown and produce no output?

Output rules

- 1 The firm sets its output where its profit is maximized.
- 2 The firm sets its output where marginal profit is equal to zero.
- 3 The firm sets its output where marginal revenue is equal to marginal cost.

Shutdown rules

- 1 The firm shuts down only if it can reduce its loss by doing so.
- 2 The firm shuts down only if its revenue is less than its avoidable cost.

Price-taking behavior

In a **competitive market**, firms are price takers.

- In both input and output markets.
- A price-taking firm faces a demand curve that is **horizontal at the market price**.
 - The individual firm faces a perfectly elastic demand curve.

The Residual Demand Curve

- Suppose Market demand is $D(p)$.
- Suppose that the (aggregate) supply curve of other firms is $S^0(p)$.
- Then the residual demand curve for an individual firm is

$$D^r(p) = D(p) - S^0(p)$$

- What are the properties of $D^r(p)$?

Competition in the Short Run & the Short Run Output Decision